Sadak Saathi

Safer roads through proactive civic action

Basic Details

PROBLEM STATEMENT

Sadak Saathi: Development of a Proactive Smart Digital Platform for Street-Level Civic Action and Road Hazard Mitigation

THEME

Smart Cities & Infrastructure

CATEGORY

Software

TEAM NAME

SheCodeX

TEAM MEMBERS

Sumedha Ghosh, Harinanda V Nair, Harini S, Yamini Bele

BST Hackathon

SheCodesX

Sadak Saathi

Sadak Saathi

Challenges and Problems

- **Hazards stay silent:** open manholes, dangling live wires, broken footpaths and waterlogging are noticed only after an accident.
- **Reporting is too heavy:** long forms and typing in English put people off, so most never report.
- **No real-time picture:** authorities cannot tell what to fix first; clerks read tickets one by one.
- **Every complaint looks equal:** a dim streetlight and an open manhole sit in the same queue.
- **Closure is not verified:** tickets are marked resolved without proof, and the problem returns.
- **Why it matters:** India recorded 1.68 lakh road deaths in 2022 (MoRTH); 39.4% were on roads other than national and state highways.

Proposed Solution

- **Photo-first, voice-first reporting:** log a hazard in seconds with a location pin, a photo or a spoken note. The app reads the photo and the voice note and fills in the hazard type.
- **Public safety map and severity queue:** every report is scored on hazard type, nearness to a school, hospital or bus stop, blocked path, night lighting, confirmations and days open. The municipal team works one ranked list.
- **Fixes verified by photograph:** the crew uploads an after-photo and residents confirm “looks fixed” or “still there”. A disputed fix goes back to the queue with a higher score.

Innovation and Uniqueness

Voice-first reporting
Hold the mic and speak in Hindi, Kannada, Tamil and more. Bhashini turns it into text.

Auto severity queue
Red, Yellow, Green tags. Live wires and open manholes jump to the top, with the score shown.

Safe-route map
A public map that routes people around open hazards. Built for night commuters and pedestrians.

Photo-verified fixes
Crew after-photo, then residents confirm “looks fixed” or “still there”.

Future Risk Map
Rainfall forecasts plus past flood and repeat-hazard reports flag streets at risk before the rain. Municipal teams get a ranked inspection list.

How it Addresses the Problems

- **Silent hazards:** a photo, pin or voice report takes seconds.
- **No real-time picture:** live public map and one ranked queue.
- **Equal priority:** severity score with Red, Yellow, Green tags.
- **Unverified closure:** after-photo plus resident confirmation.
- **Nothing back for citizens:** safe routes and early warnings.

SheCodesX

TECHNICAL APPROACH

Sadak Saathi

Methodology and Process

- **Capture:** pin, photo or voice note from a mobile web app. Reports save offline and upload later.
- **Understand:** a vision model reads the photo; Bhashini speech-to-text handles voice in the user's language.
- **Score:** rules rank severity from hazard type, nearby school, hospital or bus stop, blocked path, night light, confirmations and days open.
- **Assign:** Red, Yellow, Green queue; the right crew (electricity, sewer, roads, drainage) is suggested.
- **Verify:** crew after-photo plus resident confirmation. A disputed fix reopens with a higher score.
- **Predict:** rainfall forecast and past reports flag streets at risk of waterlogging.

Technological Stack

HTML CSS JS Leaflet BHASHINI Rajmargyatra

Your paragraph text

System Flow

Citizen Input
Photo, GPS pin or voice note. Any language, any phone.

Server (FastAPI)
Login, validation and geo-check. Merges nearby duplicates.

Data Sources
OpenStreetMap roads and landmarks, rainfall forecast, past reports.

AI Layer
Photo hazard classifier and speech-to-text (Bhashini).

Municipal Dashboard
Ranked queue, crew assignment, after-photo to close.

Severity and Risk Engine
Score, rank, Red / Yellow / Green. Future Risk Map model.

Output
Public safety map · Safe routes · Early warnings · Verified-fix feed

Offline-first: reports queue on the phone and upload when the network returns.

SheCodesX

FEASIBILITY AND VIABILITY

Sadak Saathi

Feasibility and Viability

Technical Feasibility	Financial Feasibility	Operational Feasibility	Social Viability
All parts use proven tools: OpenStreetMap, Bhashini speech, pretrained vision models, a mobile web app.	Open-source stack and free OpenStreetMap data keep running costs low.	Pilot in one ward. Ward engineers and crews use the same app.	Voice in regional languages includes people who cannot type or read English.
Photos are compressed on the phone, so reports upload fast on weak networks.	Funded by municipal budgets, smart-city funds, CSR and PPP. Free for citizens.	Exports to existing grievance portals, so current helplines stay.	A public queue and photo proof build trust.
Lightweight web app: no install, light on data, runs on older phones.	Municipalities pay a yearly platform fee per ward; CSR and grants fund the pilot.	Crews see the right job type: wire, sewer, road, drain.	Residents and RWAs confirm fixes.

Strategies for Overcoming Challenges

TECHNICAL CHALLENGES

- **Offline-first:** reports save on the phone and upload when the network returns.
- **Lightweight web app:** no install, light on data, runs on older phones.
- **Fake and duplicate control:** geo-tag check, duplicates merged, confirmations add weight.

FINANCIAL CHALLENGES

- **Phased rollout:** one ward, then the city, then the state.
- **Low running cost:** open-source tools and OpenStreetMap maps.
- **Scale-up funding:** smart-city budgets, CSR partners and PPP.

OPERATIONAL CHALLENGES

- **Training:** ward engineers and crews learn the app in one short session.
- **Integration:** link to existing grievance portals and helplines.
- **Local champions:** RWAs and student volunteers help onboard residents.

SheCodesX

IMPACT AND BENEFITS

Sadak Saathi

Citizens
- Report in seconds, in your language
- Safer routes and alerts
- Proof that reports lead to repairs

Municipal teams
- One ranked, real-time work list
- Crews get the right job and location
- Photo proof of every fix

Benefits of the solution

City and government
- Hotspot and repeat-failure analytics
- Evidence for budgets and planning
- Less manual ticket handling

Society and environment
- Fewer accidents and injuries
- Inclusive, regional-language access
- Early action on waterlogging

Potential impacts on the target audience
- **For Safety:** aims to cut injuries from open manholes, live wires and flooded roads
- **For Governance:** a ranked, real-time work list replaces manual ticket reading
- **For Citizens:** safer routes and proof that a report led to a repair
- **For Infrastructure:** streets that keep failing are found early and fixed properly
- **For Society:** inclusive reporting in regional languages
- **For Economy:** fewer accident costs, vehicle damage and lost working hours

Business Model

FUNDING SOURCES FOR SUSTAINABILITY

Municipal budgets
Ward and city funds for road safety

Urban schemes
Central and state urban development funds

CSR and grants
Corporate CSR and development agencies

SUPPORTING SOURCES

PPP
Public-private partnerships

Sponsored zones
CSR-backed safety zones

Data partners
Anonymised hazard data

TARGET CUSTOMERS
Municipal corporations and ward offices, state urban departments, road and utility agencies, and residents.

FINANCIAL MODEL
Free for citizens. Municipalities pay a yearly platform fee per ward. CSR and grants fund the pilot.

VALUE-ADDED SERVICES
Citizens: safe routes and alerts. Crews: ranked jobs and proof of work. Officials: hotspot analytics.

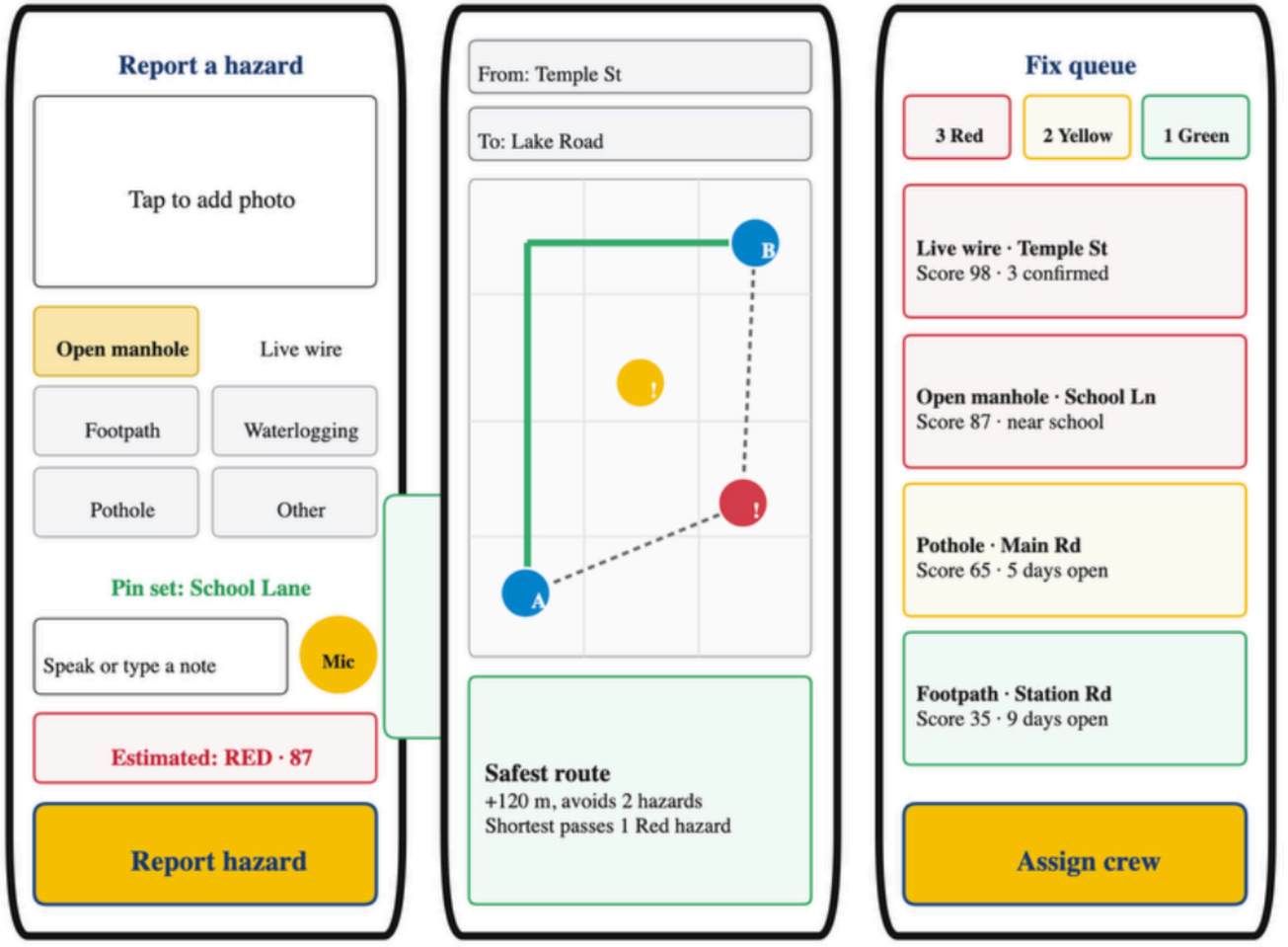
SCALABILITY
One-ward pilot, then city rollout, then state-wide through open APIs and existing grievance portals.

SheCodesX

RESEARCH AND REFERENCES

Sadak Saathi

Sample UI (concept screens)



Report in seconds | Safe-route map | Municip

Reference App & Web

[NHAI website (nhai.gov.in)]
Official portal for national highway projects, toll and complaint information. Highway-focused.

[Rajmargyatra app (NHAI)]
Highway complaints with geo-tagged photo or video, FASTag, toll and weather. Hindi and English.

[IndiaRoad Map app]
Phone-sensor pothole and speed-breaker alerts, navigation and community hazard reports.

[Safetipin app]
Crowdsourced street safety scores from nine factors such as lighting and visibility.

Our edge: street-level civic hazards (open manholes, live wires, waterlogging) with voice reporting, automatic severity ranking and photo-verified fixes.

Reference Article & Sources

- Road Accidents in India 2022 (MoRTH), via Outlook Business
- NHAI press release: Rajmargyatra app launch
- Bhashini, MeitY national language AI platform
- IndiaRoad Map app listing (developer claims)
- Safetipin, safety audits of streets (Wikipedia)